

VPA ICT — Year 6: Advanced Productivity & Web Design

Year 6: Advanced Productivity & Web Design Basics

Duration: 36 weeks (12-week units × 3) | Lesson Length: 40 minutes

Unit 1: Advanced Word Processing (Weeks 1 – 12)

Week 1 — Review & Styles

Keywords: style 样式 (yàngshì), heading 标题 (biāotí), body text 正文 (zhèngwén), template 模板 (múbǎn), table of contents 目录 (mùlù)

Activity 1 (15 min): Styles demonstration — Apply Heading 1, Heading 2, and Normal to a sample document. Watch how the Table of Contents auto-generates.

Activity 2 (15 min): Students reformat a messy document using styles only (no manual formatting).

Activity 3 (10 min): Compare: "Why are styles better than manual formatting?"

Week 2 — Mail Merge

Keywords: mail merge 邮件合并 (yóujiàn hébìng), data source 数据源 (shùjù yuán), recipient list 收件人列表 (shōujiàn rén lièbiǎo), merge field 合并域 (hébing yù)

Activity 1 (20 min): Create a class certificate mail merge. Data source = class list with names. Document = certificate template.

Activity 2 (15 min): Practice — Create personalized invitations for a school event using mail merge.

Activity 3 (5 min): Troubleshooting — What happens if the data source has a spelling mistake?

Formative: Successfully generates 10+ personalized documents.

Week 3 — Advanced Tables

Activity 1 (15 min): Create a complex table: school timetable with merged cells for lunch, colored rows for subjects, diagonal borders.

Activity 2 (15 min): Sorting data in tables — Alphabetize a class list; sort by date.

Activity 3 (10 min): Convert text to table and back.

Week 4 — Section Breaks & Columns

Keywords: section break 分节符 (fēnjié fú), column 分栏 (fēnlán), continuous 连续 (liánxù), next page 下一页 (xià yí yè), orientation 方向 (fāngxiàng)

Activity 1 (20 min): Create a newsletter: Page 1 = portrait with 2 columns; Page 2 = landscape with table.

Activity 2 (15 min): Insert section breaks to have different headers on different pages.

Activity 3 (5 min): Check understanding: "When do you need a section break vs. a page break?"

Week 5 — Review & Document Project

Project (40 min): Create a 4-page school magazine with:

Cover page with WordArt-style title

Article with 2-column layout

Interview page with formatted Q&A

Back page with class list table and mail-merge certificates

Unit 2: Advanced Spreadsheets (Weeks 6 – 12)

Week 6 — Conditional Formatting

Keywords: conditional formatting 条件格式 (tiáojiàn géshì), rule 规则 (guīzé), highlight 高亮 (gāoliàng), data bar 数据条 (shùjù tiáo), color scale 色阶 (sèjiē)

Activity 1 (15 min): Apply color scales to a gradebook — red for low, green for high.

Activity 2 (15 min): Data bars — Compare class heights with visual bars inside cells.

Activity 3 (10 min): Custom rule — Highlight all scores above 80 in green, below 60 in red.

Week 7 — IF Functions & Nested Formulas

Keywords: IF function IF函数 (IF hánshù), nested 嵌套 (qiàntào), logical test 逻辑测试 (luójí cèshì), TRUE/FALSE 真/假 (zhēn/jiǎ)

Activity 1 (15 min): Grade calculator using IF: =IF(B2>=90, "A", IF(B2>=80, "B", IF(B2>=70, "C", "D")))

Activity 2 (15 min): Attendance tracker — IF student was present, count 1; otherwise 0.

Activity 3 (10 min): Challenge — Create a shipping cost calculator: IF weight > 5kg, cost = weight * 15; else cost = weight * 10.

=IF(B2>5, B2*15, B2*10)

Week 8 — Data Validation & Protection

Keywords: data validation 数据验证 (shùjù yànzhèng), dropdown list 下拉列表 (xiàlā lièbiǎo), protect 保护 (bǎohù), password 密码 (mimǎ)

Activity 1 (15 min): Create a data entry form with dropdown lists (e.g., Class: 6A, 6B, 6C; Subject: Math, Science, English).

Activity 2 (15 min): Protect cells — Lock formulas so only input cells can be edited.

Activity 3 (10 min): Error prevention discussion — Why validate data?

Week 9 — Charts & Graphs Mastery

Activity 1 (15 min): Create a combo chart: bar for sales, line for profit on same chart.

Activity 2 (15 min): Sparklines — Mini charts inside cells showing trends.

Activity 3 (10 min): Chart critique — Analyze 3 charts: one excellent, one misleading, one confusing.

Week 10 — Introduction to HTML

Keywords: HTML 超文本标记语言 (chāo wénběn biāojiǐ yǔyán), tag 标签 (biāoqiān), element 元素 (yuánsù), attribute 属性 (shǔxìng), browser 浏览器 (liúǎnqì)

Activity 1 (15 min): "My First Webpage" — Using Notepad/TextEdit, create:

```

<html>

<body>

<h1>My Name</h1>

<p>I am in Year 6.</p>

</body>

</html>

```

Activity 2 (15 min): Experiment with h1 – h6 tags. What changes?

Activity 3 (10 min): Add a paragraph about favorite hobby.

Week 11 — CSS Basics

Keywords: CSS 层叠样式表 (céngdié yàngshì biǎo), selector 选择器 (xuǎnzéqì), property 属性 (shǔxìng), value 值 (zhí), style 样式 (yàngshì)

Activity 1 (15 min): Add inline CSS: `<h1 style="color:blue;">Hello!</h1>`

Activity 2 (15 min): Create an internal stylesheet in the `<head>` section. Style all paragraphs to be green with font-size 18px.

Activity 3 (10 min): Color experiment — Use hex colors (`#FF5733`) and RGB colors (`rgb(255,87,51)`).

```

<style>

body { background-color: lightblue; }

h1 { color: navy; text-align: center; }

p { font-family: Arial; font-size: 16px; }

</style>

```

Week 12 — Web Page Project & Assessment

Project (40 min): Create a 3-page personal website:

index.html — Homepage with name, photo, welcome message

about.html — 3 paragraphs about hobbies, family, school

contact.html — Simple form (name, email, message)

All pages must have: consistent color scheme, navigation links, and styled text.

Marking:

Bronze: 2 pages with basic HTML tags and some CSS

Silver: 3 pages with consistent styling and navigation

Gold: 3+ pages with images, tables, forms, and creative design

Standards Mapping

Unit	Standard	Description
Word Processing	CSTA 2-DA-08	Collect data using computational tools and transform it
Spreadsheets	CCSS. MATH. 6. SP. B. 5	Summarize numerical data sets in relation to context

Unit	Standard	Description
Web Design	CSTA 2-DA-09	Refine computational models based on data